



Explainable Machine Learning for WiFi Channel Occupancy and Throughput Analysis

CNG 7880 — Trustworthy and Responsible AI

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The Problem with Black-Box AI in WiFi Networks



Growing Complexity

Too many devices and signal interference make it hard for traditional models to accurately analyze WiFi networks.

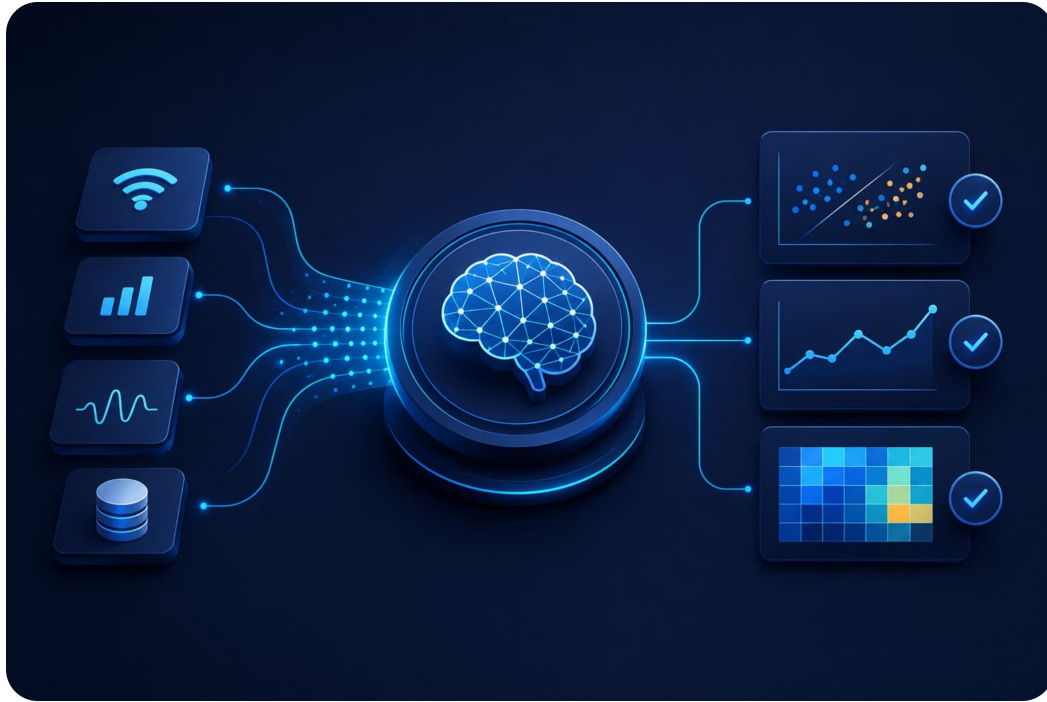
The Accuracy-Trust Gap

Modern ML achieves high accuracy but operates as a black box — decisions cannot be understood or trusted.

Real-World Consequences

A false idle prediction wastes capacity; a false busy prediction grades Quality of Service. We need models that are both **accurate** and **explainable**.

What This Study Sets Out to Do



Prediction Tasks

- **Channel Occupancy Classification** —predicting whether a WiFi channel is busy or idle at any given moment.
- **Throughput Regression** — estimating the throughput performance of that channel

Models Compared

- Random Forest, XGBoost, and LSTM with attention

Explainability Methods

- SHAP for global feature importance, LIME for local per-sample explanations, and LSTM attention heatmap(to instantly prove which WiFi channels, LSTM model is focusing on to make its decisions)

WACA Dataset – Camp Nou Stadium, Barcelona



Source & Context

Wi-Fi All-Channel Analyzer (WACA) — Barrachina-Muñoz, Bellalta & Knightly, WinTech 2020. Recorded August 2019 at Camp Nou Stadium.

Dataset Structure

- 200 iterations used from 2,001 total
- 24 WiFi channels across 4 groups (A–D), 6 sub-channels each
- 100,000 RSSI samples per channel
- 4 statistical features per channel — mean, standard deviation, maximum, and minimum — 96 features total.
- Target channel: rssi_temporal_B_b

Class Distribution

Busy threshold: RSSI mean > 30 → **140 Idle (70%), 60 Busy (30%)**

Data Preprocessing Pipeline

01

Load Raw Data

Read MATLAB `.mat` files using Python `scipy` library.

03

Threshold Calibration

Analyze RSSI distributions to confirm Busy/Idle threshold of 30.

05

Throughput Normalization

Target throughput = $\text{RSSI mean} / 1023$ *is the maximum value of the 10-bit sensor.*

02

Feature Extraction

Compute mean, std deviation, max, and min from 100,000 RSSI(Received Signal Strength Indicator) samples per channel per iteration.

04

Label Creation

Busy = 1 if mean RSSI > 30; Idle = 0 otherwise.

06

Train/Test Split

80% training (160 samples), 20% testing (40 samples) with stratification.

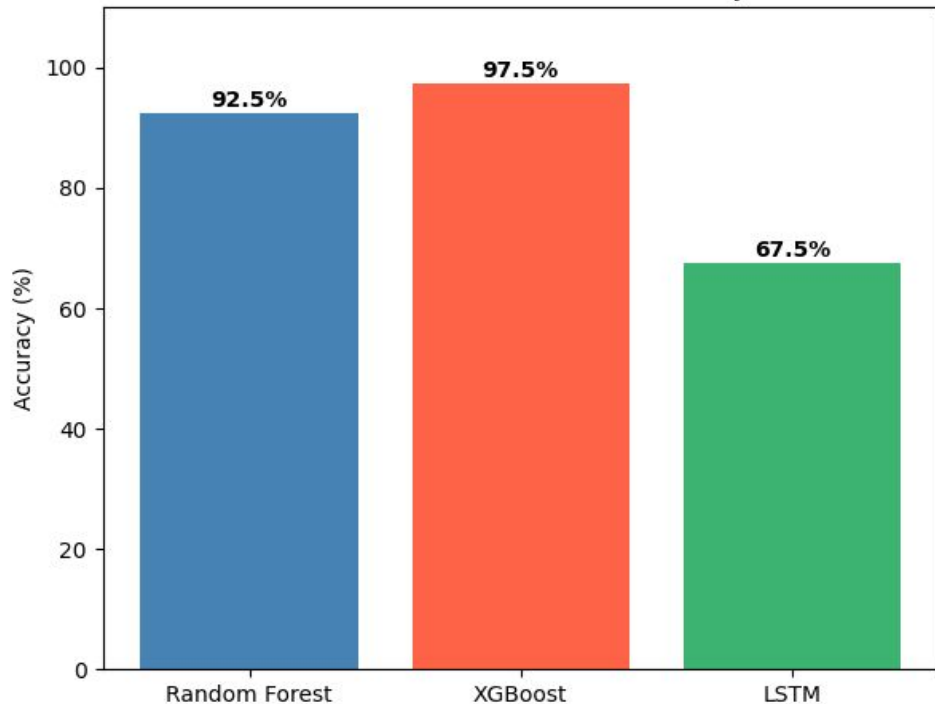
Classification & Regression Results

Model	Accuracy	F1 Score	RMSE
Random Forest	92.5%	0.857	0.0212
XGBoost	97.5%	0.957	0.0213
LSTM + Attention	67.5%	0.629	—

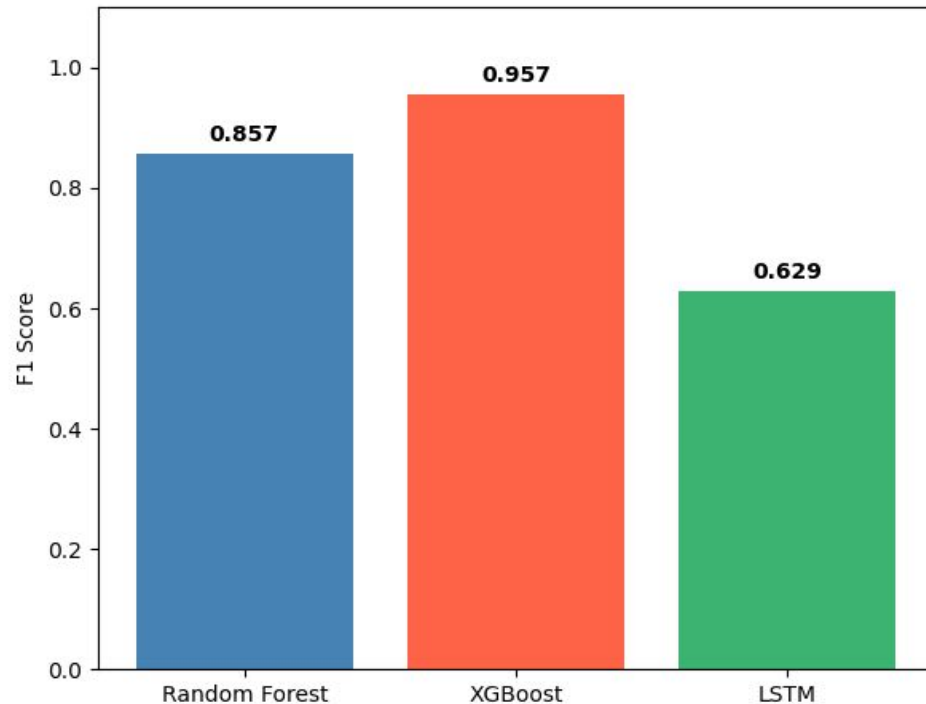
Cross-Validation & Confidence

- Random Forest: 90.5% $\pm$ 5.5% (5-fold CV), 95% CI: $\pm$ 5.6%
- XGBoost: 94.5% $\pm$ 2.6% (5-fold CV), 95% CI: $\pm$ 2.6%

All Models: Classification Accuracy

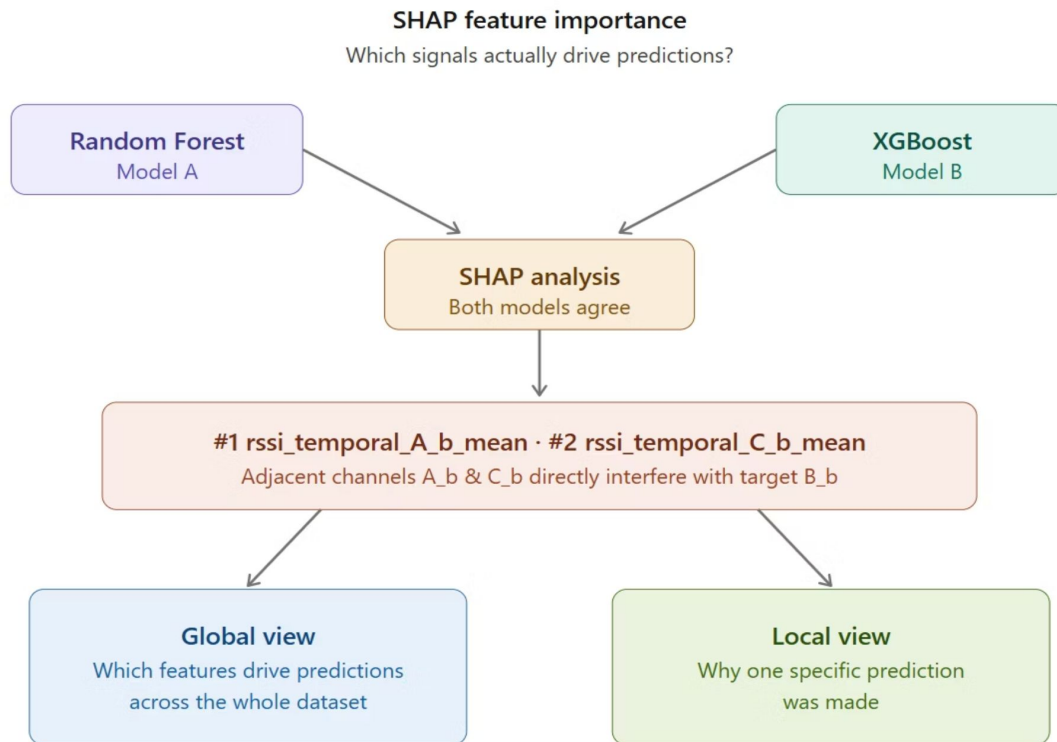


All Models: F1 Score



XGBoost achieves the highest accuracy (97.5%) and F1 score (0.957), followed by Random Forest (92.5%), while LSTM performs lower due to the tabular nature of the data.

SHAP Feature Importance – Global Explainability



LIME & LSTM Attention – Local and Temporal Explainability

LIME – Per-Prediction Explanations

LIME explains *why* the model made a specific decision for a single prediction.

- **Busy prediction:** When the neighboring channel A_b has a strong signal, the model predicts the channel is Busy
- **Idle prediction:** When neighboring channel C_b has a weak signal, the model predicts the channel is Idle
- This confirms the model is picking up on real interference patterns — not just random noise

LSTM Attention – Temporal Focus

The attention mechanism shows *which channels the model pays more attention to* over time.

- Group A channels (2.4 GHz band) get slightly more attention from the model
- All channels have fairly similar attention weights — this is expected because at a packed stadium like Camp Nou, every channel is busy at the same time
- This gives us a visual way to see inside the LSTM model — making it more trustworthy

Bias Analysis – Model Fairness Across Time

How we tested fairness: We split the 200 iterations into 3 time periods and checked if the models performed consistently — or if they favored certain time periods over others.

Early Period (*Iterations 1–66*) Very few "Busy" channel samples in this period — only 1 out of 66. This is a characteristic of the dataset (early match, less crowd activity), not a flaw in the model.

Mid Period (*Iterations 67–132*) Both models perform excellently — F1 score above 0.99. Random Forest variation: ± 0.04 , XGBoost variation: ± 0.03 .

Late Period (*Iterations 133–200*) High, consistent performance maintained across both models. No drop in accuracy detected.

Conclusion — No Bias Found Both models perform equally well across all time periods. This confirms the models are **fair, stable, and reliable** — fully meeting TRAI fairness requirements.

Thank you for your attention.

Any Question?

